

PHYS306/COSC330 Midterm 1

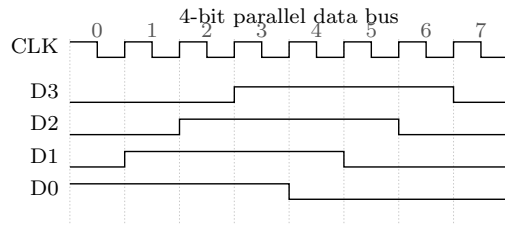
Prof. Jordan C. Hanson

Name/Student ID: _____

Instructions: Show enough work to make your reasoning clear. The total score is 100 points.

1 Chapter 1: Introductory Concepts

1. [14 points] The four traces below form a 4-bit parallel bus. Read each word as $D_3D_2D_1D_0$, with one word transferred per clock period.



The clock frequency is $f_{\text{CLK}} = 5.0 \text{ MHz}$.

- Determine the clock period.
- How much time is represented by the eight clock periods shown?
- Write the eight transmitted words as hexadecimal digits.
- Determine the total parallel bit rate.
- If the same data were sent serially at the same clock frequency, what would the bit rate be?
- What is the duty cycle of D_0 over the interval shown?

2 Chapter 2: Number Systems, Operations, and Codes

2. [18 points] Perform the following conversions and arithmetic.
- Convert 45_{10} to binary and hexadecimal.
 - Convert $0x3C$ to binary and decimal.
 - Represent -18_{10} as an 8-bit two's-complement number.
 - Treat the following as 8-bit two's-complement numbers and add them:

$$00101101 + 11110110.$$

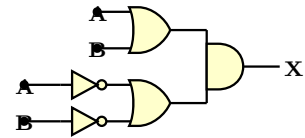
Give the 8-bit result, its decimal value, and state whether signed overflow occurs.

3. [10 points] Work with Gray code, BCD, and parity.

- Convert binary 1011 to Gray code.
- Convert Gray code 1110 to binary.
- Interpret the BCD word 0010 0101 as a decimal number.
- A transmitted word uses one leading even-parity bit followed by two BCD digits. The receiver obtains 0 0010 0101. Has a parity error been detected? Explain using the number of HIGH bits.

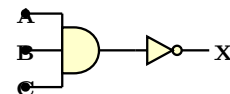
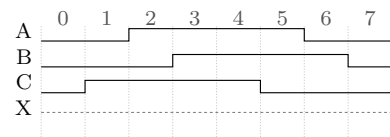
3 Chapter 3: Logic Gates

4. [12 points] Consider the gate network below.



- Write the Boolean expression for X .
- Construct the complete truth table.
- Simplify the expression using Boolean algebra.
- Which single basic logic gate performs the same function?

5. [10 points] The data below are inputs to the gate shown.



- Draw the output waveform for X .
- If the bit interval is 20 ns, give the width of the LOW pulse in X .

6. [12 points] NAND gates are universal logic elements.
- (a) Using only 2-input NAND gates, draw a circuit that implements a 2-input OR gate.
 - (b) We apply $A = 1$ and $B = 0$ to a 2-input NAND gate, and the output is $X = 0$. Is that gate behaving correctly? Why or why not?

4 Chapter 4: Boolean Algebra and Logic Simplification

7. [14 points] Simplify the following logic functions.

- (a) Apply DeMorgan's theorems to

$$Y = \overline{(A + \overline{B})(C + \overline{D})}.$$

- (b) Use Boolean algebra to reduce

$$Z = A\overline{B} + AB + \overline{A}B$$

to its minimum form.

- (c) Use a 4-variable Karnaugh map to find the minimum SOP form of

$$\begin{aligned} X = & \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}CD + \overline{A}B\overline{C}D + \overline{A}BCD \\ & + AB\overline{C}\overline{D} + AB\overline{C}D + ABC\overline{D} + ABCD. \end{aligned}$$

Draw the minimum circuit using basic gates.

5 Chapter 5: Combinational Logic Analysis

8. [10 points] A receiver accepts three data bits A, B, C and a received parity bit P . The transmitter uses **odd parity**. Define an error output E so that $E = 0$ when the received four-bit word has correct odd parity, and $E = 1$ when the parity is incorrect.
- (a) Write a Boolean expression for E using XOR/XNOR notation.
 - (b) For $ABC = 101$, determine the parity bit P that should be transmitted.
 - (c) If $ABC = 101$ is received with $P = 0$, determine E .
 - (d) Draw the minimum gate-level circuit for the parity checker.